Paddy Land Watch for Sri Lanka

Oct 21, 2026

At a glance

Every Yala and Maha season, Paddy Land Watch tells the Department of Agrarian Development which paddy tracts grew paddy, which grew another crop, which lay idle and which were filled in. It works from free Sentinel-1 radar and Sentinel-2 images, read by deep learning.

Item	Detail
Question it answers	What happened on each paddy tract this season, and where is new filling?
Study area	2–3 wet-zone districts near Moratuwa (Gampaha, Colombo, Kalutara), where cloud blocks optical images in the monsoons
Users	Department of Agrarian Development and its Agrarian Service Centres; Department of Census and Statistics
Output	A four-state map per tract each season, plus a ranked list of likely filling cases and idle tracts to inspect
Cost	Free data and software (Google Earth Engine, Python); field travel only
Team and time	3 members, October 2026 – June 2027; Maha 2026/27 is the live season for field checks

The problem

By law, paddy land must be farmed every season, yet a lot of it lies idle or is filled in, and the department finds out late, on paper.

- **The law:** the Agrarian Development Act No. 46 of 2000 says paddy land "shall be cultivated with paddy during every season" (s. 30). Filling it without the Commissioner-General's written permission is an offence, with a fine of up to Rs 100,000 or up to six months in prison (s. 32).

- **Idle land is large:** in February 2025 the department's "Eka Mitata Govi Bimata" survey found over 100,000 acres of uncultivated paddy fields. The report gives no districts and no survey method.
- **Money depends on the land state:** for Yala 2026 the state pays Rs 25,000 per hectare for paddy and Rs 15,000 per hectare for other crops on paddy land where paddy is not grown, up to 2 ha per farmer.
- **Filling slips through:** a 2018 Auditor General special audit of Kegalle District found 63 acres 2 roods 12 perches filled in 372 instances. Cultivated paddy had fallen 76% since 1990, and no legal action had been taken.
- **The records are on paper:** the same audit describes an Agricultural Land Register and a Fallow Paddy Land Register, and none of its recommendations mention maps, satellites or software.

The key idea

Each of the four land states leaves its own pattern across a season, and radar records that pattern even under monsoon cloud.

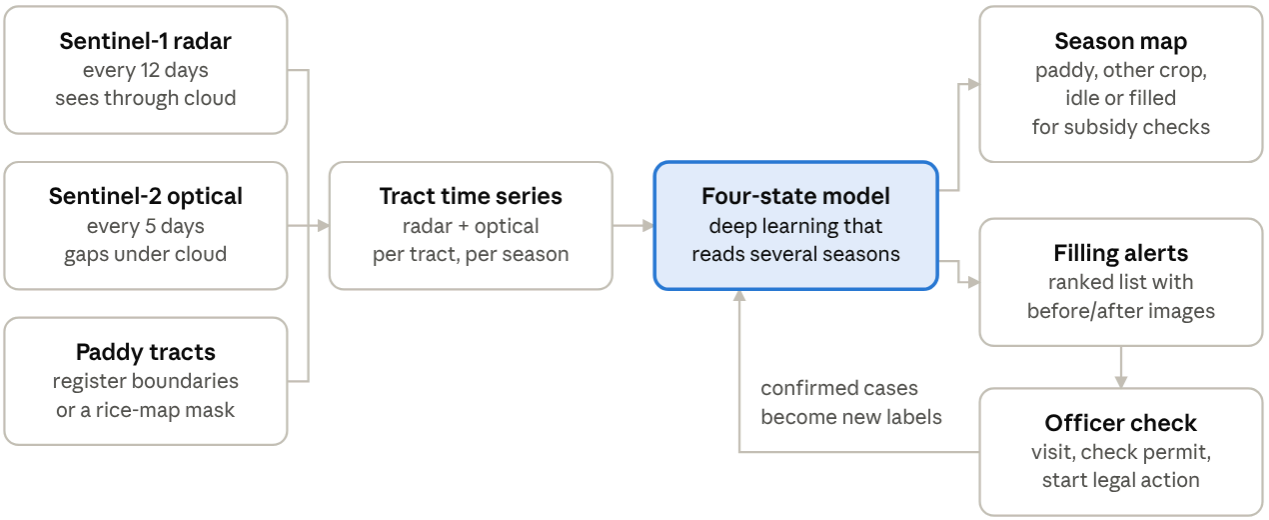
Land state	What the season looks like from space	Why the department cares
Paddy grown	The field floods at planting (radar goes dark), the crop greens up, then it is harvested	Paddy subsidy, Rs 25,000/ha
Other crop on paddy land	No standing water; a shorter green-up from maize, vegetables or similar	Other-crop subsidy, Rs 15,000/ha
Idle	No flooding and no crop cycle; weeds or grass give a flat signal	Target for idle-land recovery
Filled or built over	A sudden, lasting change to bare soil or roofs; the field never floods again	Possible offence under s. 32; inspect quickly

The AI is needed because wet-zone tracts are small and mixed, and cloud hides the optical signal. The national rice maps of Özdoğan et al. (2025) say western and central districts "are expected to have lower accuracies", and call radar the obvious approach "in such cloudy environments". Their own early test, though, found Sentinel-1 errors about equal to optical ones, so this project tests radar against optical-only directly (RQ3).

A model that reads several seasons at once can also tell a one-off fallow season from land that has been abandoned.

How it works

Free radar and optical images become a season map and a ranked alert list



Paddy Land Watch pipeline · 9 steps, 1 feedback loop

Radar keeps the model working through cloudy weeks. Each case an officer confirms becomes a new training label, so the model can be retrained every season.

What is new

Sri Lankan paddy has been mapped before, but only as "rice or not rice". Nobody we found tracks idle, other-crop and filled paddy land each season, or sends alerts.

Study	Where	What it did	What it leaves out
Özdoğan et al. 2025, Remote Sensing	All Sri Lanka, Maha and Yala 2000–2022	Rice area and yield from Landsat and Sentinel-2; about 91–93% accuracy in 2021 (70% to over 90% across years)	Optical only; other crops lumped as "non-rice"; no idle or filled class; no field boundaries
Sankavi et al. 2025, University of Moratuwa	North Central Province	CNN finds paddy; NDVI/NDWI split active and inactive fields	Dry zone; no other-crop or filled class

Study	Where	What it did	What it leaves out
Wanninayaka et al. 2020, KDU	Ampara	Sentinel-1 time series maps paddy area (0.92 threshold, 0.86 random forest)	Paddy area only
Jayasooriya et al. 2023, KDU	Hambantota, near the Southern Expressway	Landsat 1990–2022 shows paddy lost near the road	Looks back; no alerts
Geo-Goviya, IWMI and the Agriculture Ministry, 2023	National	Information system with land use, paddy area and area planted per season	No idle or filled land mentioned
Kasture et al. 2025, EGU abstract	Udupi, India	Sentinel-1 radar with LSTM/BiGRU models detects abandoned paddy	Closest method; two classes only; no accuracy given; not Sri Lanka

Our gap, worded carefully: to our knowledge, no study classifies Sri Lankan paddy land every season into the four states the law and the subsidy use (paddy, other crop, idle, filled). None does it in the cloudy wet zone by fusing Sentinel-1 and Sentinel-2 with deep learning, or tests the result as an alert list the department checks in the field.

Data and ground truth

All imagery is free; the only data that must be asked for is the department's own lists, so that request goes out in week one.

What	Source	Note
Radar time series	Sentinel-1, 10 m, through cloud, in Google Earth Engine	Planned every 6 days, but Gampaha got 9 scenes in Jul–Sep 2026, mostly 12 days apart (Copernicus)
Optical time series	Sentinel-2, 10 m, about every 5 days	Many gaps under monsoon cloud
Paddy mask and baseline	Özdoğan et al. seasonal rice maps and code (GitHub)	Shows which land has grown rice since 2000

What	Source	Note
Few-label baseline	AlphaEarth Satellite Embedding, 10 m, 2017–2024, CC-BY 4.0 (catalogue)	Tests whether a simple model on embeddings matches deep learning
District totals	Sown and harvested extent by district, Maha and Yala (Census and Statistics)	A check on totals, not on tracts
Field labels	Team visits in Gampaha and Kalutara during Maha 2026/27: GPS phone photos of about 400 tract-seasons	The live season falls inside the project
Department records	Eka Mitata uncultivated-land list, Fallow Paddy Land Registers at Agrarian Service Centres, filling permits	Not public; request with a supervisor's letter
Stretch input	NISAR L-band radar, provisional products since 20 July 2026 (ASF)	Does L-band separate idle land from other crops better?

Labels come from field visits and from Sentinel-2 itself, not traced from Google imagery: Google's terms for its map content say "You will not create content based on Google Maps Content" (terms, via Cesium).

Who uses it

The department already has field officers and a digital paddy register; what it lacks is a quick way to see change. Its 2023 performance report lists 565 Agrarian Service Centres under 25 district offices. It names "the excessive demand for filling up of paddy lands" as a challenge and legal action against violators as a function.

User	What they get	What they do with it
Agrarian Service Centre officers	A monthly ranked list of likely filling cases, with before-and-after image chips	Visit the top cases, check the permit, start legal action where none exists
Idle-land recovery programme	Tracts idle for two or more seasons, by Service Centre	Target owners; the 2023 report says 556 acres (Yala) and 2,429 acres (Maha) were brought back

User	What they get	What they do with it
Subsidy officers	Each tract's state this season: paddy, other crop or idle	Spot-check Rs 25,000 and Rs 15,000 per hectare claims
Department of Census and Statistics	An independent sown-area estimate per district	Compare with survey-based figures

The output is an alert to check, never a verdict. In January 2021, a paddy field in Kesbewa, next to Moratuwa, was filled with building debris while councillors argued over whether it had permits. A satellite sees the change; only an officer can say whether it was legal.

The department's register fits in directly: the report says its electronic paddy land register is complete and boundaries are being mapped with the Geo-Goviya app. Those boundaries would replace hand-drawn tracts.

Research questions and team roles

Four questions, each with one owner, so the work splits cleanly across three people.

RQ	Question	Owner
RQ1	How accurately can deep learning on fused Sentinel-1 and Sentinel-2 time series sort each tract's season into paddy, other crop, idle or filled, in the cloudy wet zone?	A
RQ2	How soon after filling starts can change detection flag it, and how many of the top 20 and top 50 alerts are real?	B
RQ3	Does reading several seasons at once separate a one-off fallow from abandonment, and does radar hold accuracy when clear optical images are few?	A, B
RQ4	Which way of ranking alerts (by area, speed of change or nearness to waterways) gives the most confirmed cases per field visit?	C
Stretch	Does NISAR L-band radar separate idle land from other crops better than Sentinel-1?	B

- **A: land-state model.** Time-series deep learning (for example a temporal CNN or transformer) and its baselines.
- **B: change and validation.** Filling detection, field labels, accuracy and alert checks.

- **C: pipeline and decisions.** Earth Engine to model to alerts, the dashboard, alert ranking and the hand-off to the department.

How we prove it

Every result is tested against field labels the model never saw, and against a simpler method it has to beat.

Tested	Metric	Against
Land-state map (RQ1)	F1 per class and macro-F1, tested in held-out areas	Random Forest on the same inputs; AlphaEarth embeddings with a simple classifier; Özdoğan rice maps for the paddy class
Cloudy seasons (RQ3)	Macro-F1 when clear Sentinel-2 images are few, reported separately	The same model with optical data only
Fallow vs abandoned (RQ3)	Accuracy on tracts labelled for two or more seasons	A model that sees one season only
Filling alerts (RQ2)	Share of the top 20 and top 50 alerts confirmed in the field; days from start of filling to alert	Simple before-and-after change in radar and NDVI
District totals	Percentage difference in sown paddy area per district	Census and Statistics seasonal figures
Alert ranking (RQ4)	Confirmed cases per field visit	Ranking by area only

Scope, risks and the first six weeks

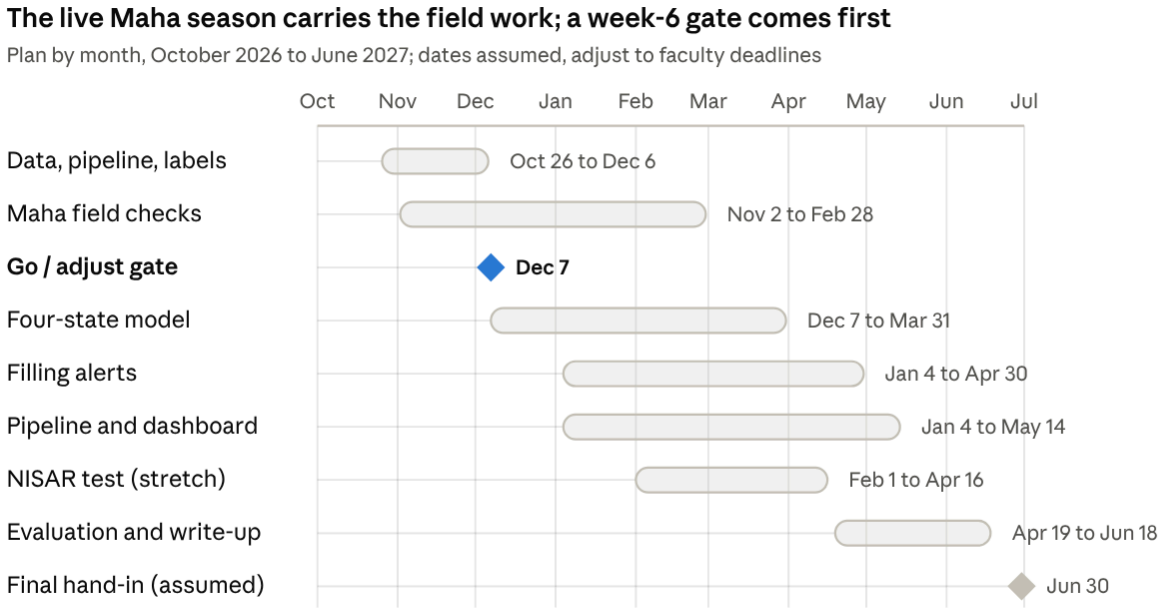
The project stands or falls on two things: separable classes and some department data. Both get tested before mid-December.

- **Core:** Gampaha and Kalutara, seasons from Maha 2022/23 to Maha 2026/27, the four-state model, filling alerts, validation and a dashboard.
- **Stretch:** the NISAR L-band test; one dry-zone district to test transfer; a subsidy-check view.
- **Out:** the whole island, legal rulings, drones, a mobile app, yield estimates.

Risk	Early sign	Fallback
The department does not share its lists	No reply by week 3	Field labels only; ask one Agrarian Service Centre to check 50 alerts
Idle and other-crop tracts look alike	F1 below about 0.6 for those two classes	Merge them into "not paddy" and keep filling as the main result
Someone already built this	An unidentified "SL-Paddy" web app turned up in a search but would not load	Find its owner in week 1; if it overlaps, build on it rather than repeat it
Legal filling looks like illegal filling	Permitted sites appear among alerts	Alerts only; officers check permits

1. **Week 1:** find the owner of "SL-Paddy"; read the full Sankavi et al. paper; send the data request (Eka Mitata list, fallow registers, filling permits, register boundaries).
2. **Week 2:** Earth Engine pipeline for one Divisional Secretariat area; paddy mask from the Özdoğan maps.
3. **Week 3:** label about 400 tract-seasons in the field and on Sentinel-2; collect 30–50 known filling cases.
4. **Week 4:** Random Forest baseline, scored against the Özdoğan maps on the same labels.
5. **Week 5:** first deep-learning model against the baseline; check how many of the top 20 filling alerts are real.
6. **Week 6:** go or adjust: macro-F1 of 0.70 or more, at least one department data source or officer check, and no live system already doing this.

Plan



Project plan · October 2026 to June 2027

Field checks run through Maha 2026/27, so labels come from a live season. The week-6 gate in early December decides whether to merge classes or narrow the scope before the model work starts.

How it compares with the SWOT and roadside projects

All three answer a question a named agency asks; they differ in where the risk sits. Paddy Land Watch carries the least technical risk, but its novelty comes from new classes and alerts, not a new sensor.

Item	SWOT Tank Water Tracker	Roadside Tree & Power-Line Risk	Paddy Land Watch
Question	How much water does each tank hold right now?	Which roadside trees could fall, or are growing into the wires?	What happened on each paddy tract this season, and where is new filling?
Main users	Irrigation staff, farmer organisations	Councils, RDA, CEB, LECO	Department of Agrarian Development, Census and Statistics
Data	SWOT heights plus Sentinel-1/2	Phone video from a moving vehicle	Sentinel-1/2 time series

Item	SWOT Tank Water Tracker	Roadside Tree & Power-Line Risk	Paddy Land Watch
AI core	Water maps under cloud and floating plants; storage curves	Tree and cable detection; 3D wire triangulation	Time-series land-state model; change detection
What is new	First SWOT use in Sri Lanka; first satellite tank volumes here	No study found measuring branch-to-wire gaps from phone video	No seasonal four-state paddy map or filling alerts in Sri Lanka
Ground truth	Irrigation Department gauge readings	Two-point laser measurements, expert risk ratings	Field visits, department lists
Field work	A few trips to tanks	Filming about 100 km of roads, laser checks	Visits to paddy tracts near Moratuwa
Biggest risk	Small tanks below SWOT's 6.25 ha accuracy target	Gap accuracy, live wires, privacy	Idle versus other-crop separation; department data
Kit	Free data; travel	Phones, damped mount, rangefinder	Free data; travel

Sources

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- Ada Derana, 4 Mar 2026: fertilizer subsidies for Yala 2026
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- Department of Agrarian Development, Annual Performance Report 2023
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- Department of Census and Statistics: paddy statistics
- ASF: NISAR data availability
- Google Maps content terms, via Cesium